Geometric Type Theory

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1 Introduction

2 Axioms

1. There is a topos $\mathcal{B}$. **Remark**: It would be better to say there is a logos $\mathcal{B}$. We should imagine $\mathcal{B}$ comes with algebraic operations; colim and lim. with corresponding algebraic laws. Some difficulty with this includes presentability.. Maybe this doesn't matter so much internally?
2. TODO: Find a property of $\mathcal{B}$ corresponding to the étale topology
3. For every (finitely/countably) presented geometric theory $\mathbb{T}$, the natural map

$$\mathcal{B}[\mathbb{T}] \to (\mathrm{Geom}_{\mathcal{B}}(\mathcal{B}[\mathbb{T}], \mathcal{B}) \to \mathcal{B})$$

   is an equivalence.
4. Étale locale choice.

We introduce notation $\mathrm{Sh}(\mathcal{L}) := \mathrm{Log}_{\mathcal{B}}(\mathcal{L}, \mathcal{B})$. This corresponds to our algebraic/geometric intuition. $\mathcal{B}$ is our base logos, and $\mathrm{Sh}(\mathcal{B}) = 1$. $\mathcal{B}[X]$ is the logos of sets, and $\mathrm{Sh}(\mathcal{B}[X]) = \mathcal{B}$.

3 Realisation

Given an object $x : \mathcal{B}$ there is a realisation $T(x)$ given by $\hom(1, x)$. For instance, take the natural numbers object $N : \mathcal{B}$. Then we have a map $\mathbb{N} \to T(N)$ given by, for each n taking the internal successor of zero n-times.

We have a universe $\mathcal{U}_{\mathcal{B}} := \Sigma_{x:\mathcal{B}} \hom(1, x)$.

Lemma 3.1 *The universe $\mathcal{U}_{\mathcal{B}}$ is closed under Π, Σ, $=$, limits/colimits, $\mathbb{N}$.*

More generally given a logos $\mathcal{B} \to \mathcal{L}$ and an $x : \mathcal{L}$ we have a realisation

4 Simplicial aspects

There are several options for what could be used as an "interval". There is $\mathcal{S} := [\mathbf{2}, \mathcal{B}]$ the "sierpinski" topos. $\mathcal{B}$-logos morphisms from $Sh(\mathcal{S}) := \mathcal{S} \to \mathcal{B}$ correspond to objects of $\mathcal{B}$ with at most 1 element. That is $\mathrm{Sh}(\mathcal{S}) = \Omega \subseteq \mathcal{B}$ is the geometric propositions.